# Transport in plants

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

## 1 · Part 1

Core | Xylem and phloem | [1]

**Answer: B**

### Explanation

Xylem transports water and mineral ions; phloem transports sucrose and amino acids. Option A puts sucrose in xylem and water in phloem, option C swaps amino acids and mineral ions, and option D reverses the two tissues completely.

## 2 · Part 1

Extended | Xylem and phloem | [1]

**Answer: D**

### Explanation

Xylem cells are joined end to end with no cross walls, which forms a long continuous tube. Lignin (option A) makes the walls thick and strong but does not make the tube continuous. Xylem vessels have no cell contents, so they have no mitochondria (option B). Sieve plates (option C) are found in phloem, not xylem.

## 3 · Part 1

Core | Water uptake | [1]

**Answer: A**

### Explanation

Water enters root hair cells from the soil, passes through the root cortex cells into the xylem, and the xylem carries it up the stem to the mesophyll cells of the leaf. Option B starts in the wrong cells, option C puts the cortex after the xylem, and option D uses phloem, which does not carry water up from the roots.

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## 4 · Part 1 Core Water uptake [1]

**Answer: C**

### Explanation

Root hairs give the root a large surface area for the uptake of water and mineral ions, so losing them reduces uptake. Option A is wrong because the roots cannot absorb more without their root hairs, whatever the volume of soil. Option B is wrong because less water enters the root, so less reaches the xylem. Option D is wrong because the cortex cells have a much smaller surface area than the root hairs that were lost.

## 5 · Part 1 Core Transpiration [1]

**Answer: D**

### Explanation

Water evaporates from the surfaces of the mesophyll cells into the air spaces and then diffuses out through the stomata as water vapour. Transpiration is the loss of water vapour, not liquid water (options A and B), it is not active transport (option B), and the waxy cuticle reduces water loss rather than letting vapour pass out directly (option C).

## 6 · Part 1 Core Transpiration [1]

**Answer: A**

### Worked solution

Compare one factor at a time. Temperature with still air: 12 mm at 15 °C and 21 mm at 25 °C, an increase. Temperature with the fan: 26 mm and 40 mm, an increase. Wind with 15 °C: 12 mm still and 26 mm with the fan, an increase. Wind with 25 °C: 21 mm and 40 mm, an increase. Both factors increase transpiration.

## 7 · Part 1 Extended Transpiration [1]

**Answer: C**

### Explanation

Water is pulled up the xylem by the transpiration pull, which works because forces of attraction between water molecules hold them together in a continuous column. An air gap breaks the column, so the pull cannot be passed to the water below the gap. Option A is wrong because the bubble is inside one vessel and transpiration continues. Option B is wrong because water does not move up xylem by active transport of ions. Option D is wrong because air does not remove lignin.

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## 8 · Part 1 Extended Transpiration [1]

**Answer: B**

### Explanation

Higher humidity means more water vapour in the air outside the leaf, so the concentration gradient for diffusion of water vapour out of the stomata is smaller and transpiration slows. Options A, C and D all describe changes that increase the rate of transpiration, so none of them is a decrease.

## 9 · Part 1 Extended Translocation [1]

**Answer: A**

### Explanation

Translocation is the movement of sucrose and amino acids in phloem from sources (parts that release them) to sinks (parts that use or store them). Option B describes water transport in xylem, option C names the wrong tissue and the wrong direction, and option D describes transpiration.

## 10 · Part 1 Extended Translocation [1]

**Answer: D**

### Worked solution

Ask at each time: does this part release sucrose (source) or use or store it (sink)? Summer: the tuber is filling up with stored food, so it is a sink. Spring: the tuber is emptying its store to feed growing shoots, so it is a source. The same organ can be a sink at one time and a source at another.

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## 11 · Part 1 Extended Xylem and phloem [6]

**Mark scheme**

**(a)** **[1]**

- sucrose and amino acids (1)

**REJECT** glucose or sugar instead of sucrose

**IGNORE** water

**(b)** **[1]**

- support (for the plant or stem) (1)

**ALLOW** strength or holding the plant upright

**(c)** **[4]**

- thick walls (1)
- containing lignin (1)
- make the vessels strong so they do not collapse, or give support (1)
- no cell contents (1)
- so there is nothing to slow down or block the flow of water (1)
- cells joined end to end with no cross walls (1)
- forming a long continuous tube (so water can flow freely from roots to leaves) (1)
- max 4; at least one mark must link a feature to its function to score full marks

**Worked solution**

(a) Phloem carries the products of the leaves: sucrose and amino acids. (b) Xylem has two jobs: transporting water and mineral ions, and supporting the plant. (c) Pair each feature with what it does. Thick walls strengthened with lignin stop the vessels collapsing and help hold the plant up. Xylem vessels have no cell contents, so there is nothing inside them to get in the way of water. The cells are joined end to end with no cross walls, so they form one long continuous tube from the roots to the leaves. An "explain" answer needs the reason, not just the list of features.

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## 12 · Part 1 Core Water uptake [6]

**Mark scheme**

**(a)** **[1]**

- so that the pathway of the water can be seen (because the dye moves with the water) (1)

**ALLOW** to show where the water goes or how far it travels

**(b)** **[1]**

- xylem (1)

**REJECT** phloem

**(c)** **[2]**

- 90 ÷ 60 (1)
- 1.5 mm per min (1)

**ALLOW** 1.5 without a unit for 2 marks (the unit is given in the question)

**(d)** **[2]**

- the blue colour would travel more slowly or a shorter distance in the same time (1)
- because there are no leaves, so (much) less water is lost by transpiration or evaporation, so less water is drawn up the stalk (1)

**ALLOW** no leaves means no stomata for water vapour to leave through

**Worked solution**

(a) Water is colourless, so you cannot see where it goes. The dye travels with the water and shows its path. (b) The tissue that carries water up the stem is xylem, so that is where the dye appears. (c) Mean rate = distance ÷ time = 90 mm ÷ 60 min = 1.5 mm per min. Check: 1.5 mm per min for 60 min gives 90 mm. (d) Most water is lost from the leaves. Without leaves, very little water is lost from the shoot, so much less water moves up the stalk and the dye rises more slowly.

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## 13 · Part 1 Core Water uptake [5]

**Mark scheme**

**(a)** **[2]**

- Y (1)
- larger surface area of root hairs (four times larger) for absorption or uptake of water (1)
- second mark depends on choosing Y

**(b)** **[3]**

- (water enters) root hair cells (1)
- then (passes through) root cortex cells (1)
- then (enters and travels up) the xylem (to the leaf) (1)
- order must be correct for full marks; deduct one mark for each item out of order

**REJECT** phloem in place of xylem

**Worked solution**

(a) Root hairs increase the surface area of the root, and a larger surface area increases the uptake of water and mineral ions. Seedling Y has 600 $mm^2$ of root hair surface compared with 150 $mm^2$ for X, which is four times as much, so Y should take up water more quickly. (b) Water from the soil enters the root hair cells first. It moves across the root cortex cells to the xylem in the centre of the root. The xylem carries it up the stem and into the leaf, where it reaches the mesophyll cells.

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## 14 · Part 1 Core Transpiration [8]

**Mark scheme**

**(a)** **[2]**

- 47 ÷ 5 (1)
- 9.4 mm per min (1)

**(b)** **[2]**

- as wind speed increases, the rate of transpiration increases (1)
- the increase becomes smaller at the highest wind speed or between settings 2 and 3, or a correct comparison using data from the table (1)

**ALLOW** rate increases nearly four times from setting 0 to setting 3

**(c)** **[1]**

- some of the water taken up is used by the plant (e.g. in photosynthesis or to keep cells turgid) and is not lost as vapour (1)

**ALLOW** some water is stored in the cells of the shoot

**(d)** **[3]**

- carry out the experiment at several (at least three) different temperatures, e.g. using a heater or rooms or chambers at different temperatures (1)
- keep the fan at the same setting (or off) and the same shoot, light intensity and humidity (1)
- measure the distance the bubble moves in the same time (5 minutes) at each temperature, and repeat for a mean (1)

**Worked solution**

(a) Rate = distance ÷ time = 47 mm ÷ 5 min = 9.4 mm per min. Check: 9.4 × 5 = 47. (b) The distances rise from 14 mm to 31 mm to 47 mm to 55 mm, so faster air movement increases transpiration. The increases are 17 mm, 16 mm and then only 8 mm, so the effect gets smaller at the highest wind speed. (c) A potometer measures how much water goes into the shoot. Most of it is lost by transpiration, but a little is used by the cells, for example in photosynthesis. (d) Change only the temperature: use at least three different temperatures, keep everything else the same (fan setting, shoot, light, humidity), and measure how far the bubble moves in the same time at each one. Repeating each reading and calculating a mean makes the results more reliable.

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## 15 · Part 1 Extended Transpiration [6]

**Mark scheme**

**(a)** **[1]**

- from the surfaces of the mesophyll cells (into the air spaces) (1)

**REJECT** from the stomata or guard cells

**(b)** **[3]**

- P (1)
- P has more stomata per $mm^2$ (250 compared with 90) (1)
- P has wider stomata (8 µm compared with 3 µm), so more water vapour can diffuse out of the leaf through the stomata (1)
- marks 2 and 3 depend on choosing P

**(c)** **[2]**

- the air spaces give a large internal surface area (of mesophyll cells) (1)
- so more water can evaporate from the cell surfaces into the air spaces, so more water vapour is available to diffuse out of the stomata (1)

**Worked solution**

(a) Water evaporates from the wet surfaces of the mesophyll cells into the air spaces inside the leaf. (b) Water vapour leaves the leaf by diffusing through the stomata. Species P has almost three times as many stomata per $mm^2$ and each open stoma is wider, so there are more and larger openings for the vapour to diffuse through. P will lose water vapour faster. (c) Because the air spaces are interconnected, many mesophyll cell surfaces are exposed to air. This large internal surface area means a lot of water evaporates into the air spaces, so there is plenty of water vapour to diffuse out through the stomata.

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## 16 · Part 1 Extended Transpiration [6]

**Mark scheme**

**(a)** **[4]**

- water vapour is lost from the leaves by transpiration (1)
- this creates a transpiration pull (1)
- which draws water up the xylem (from the roots) (1)
- water molecules are held together by forces of attraction (between water molecules) (1)
- so water moves up as a continuous column (1)
- max 4

**(b)** **[2]**

- no leaves, so very little transpiration or water vapour loss (1)
- so (almost) no transpiration pull to draw water up the xylem (1)

**Worked solution**

(a) Water evaporates from the mesophyll cells and leaves as water vapour through the stomata. This loss creates a transpiration pull at the top of the xylem. Because water molecules are attracted to each other, they stay together as a continuous column, so when molecules at the top are pulled, the whole column is drawn up the xylem from the roots. (b) Transpiration happens mainly from leaves. In winter the oak has no leaves, so there is very little water loss and so very little transpiration pull. Without the pull, water hardly moves up the xylem.

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## 17 · Part 1 Extended Transpiration [7]

**Mark scheme**

**(a)** **[2]**

- (9.6 − 3.0) ÷ 9.6 × 100 (1)
- 69% (1)

**ALLOW** 68.75% or 68.8%

**(b)** **[3]**

- higher humidity means more water vapour in the air outside the leaf (1)
- so the concentration gradient of water vapour between the air spaces and the outside air is smaller (1)
- so water vapour diffuses out of the stomata more slowly or the rate of transpiration decreases (1)

**(c)** **[2]**

- higher temperature (30 °C compared with 20 °C) gives water molecules more kinetic energy (1)
- so water evaporates faster from the mesophyll cell surfaces or water vapour diffuses out faster (1)

**Worked solution**

(a) At 20 °C, the mass lost fell from 9.6 g to 3.0 g. Decrease = 9.6 − 3.0 = 6.6 g. Percentage decrease = 6.6 ÷ 9.6 × 100 = 68.75%, which rounds to 69%. Check: 31.25% of 9.6 g is 3.0 g, and 100% − 31.25% = 68.75%. (b) Water vapour diffuses out of the leaf from where there is more of it (the air spaces) to where there is less (the outside air). When the outside air is humid, the difference is smaller, so diffusion is slower and less water is lost. (c) Plants 1 and 3 had the same humidity, so the difference is caused by temperature. At a higher temperature water molecules have more kinetic energy, so more water evaporates from the mesophyll cells and the vapour diffuses out faster.

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## 18 · Part 1 Extended Transpiration [5]

**Mark scheme**

**(a)** **[1]**

- wilting (1)

**(b)** **[4]**

- water is lost by transpiration faster than it is taken up by the roots (1)
- (because it is hot and dry and) the soil is dry, so little water is available to absorb (1)
- (leaf and stem) cells lose water (by osmosis) (1)
- cells become flaccid or lose turgor (pressure) (1)
- cells no longer press against each other, so the leaves and stems are no longer supported (1)
- max 4

**Worked solution**

(a) The drooping is called wilting. (b) On a hot, dry afternoon the plant loses water from its leaves by transpiration faster than its roots can take water from the dry soil. The cells of the leaves and young stems lose water by osmosis and become flaccid instead of turgid. Soft stems and leaves are held up mainly by turgid cells pressing against each other, so when the cells become flaccid the leaves and stems droop. After watering, the cells take up water, become turgid again, and the plant stands upright.

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## 19 · Part 1 Extended Translocation [6]

**Mark scheme**

**(a)** **[1]**

- a part of a plant that uses or stores sucrose or amino acids (1)

**(b)** **[1]**

- phloem (1)

**(c)** **[4]**

- in the first year the leaves photosynthesise and are the source (1)
- the root receives sucrose and stores it, so it is a sink (1)
- in the second year (early spring) the root releases the stored sucrose, so it is a source (1)
- the sucrose is translocated to the growing flowering shoot or flowers, which are sinks (1)
- because the new shoot does not yet have enough leaves to make its own food (1)
- max 4

**Worked solution**

(a) A sink is any part of a plant that uses or stores sucrose or amino acids. (b) Sucrose is translocated in the phloem. (c) A source releases sucrose and a sink uses or stores it. In the first year, sucrose made in the leaves is translocated down to the root and stored there, so the root is a sink. In the second year, the new flowering shoot grows quickly early in spring, before it has enough leaves to supply itself. The root now releases its stored sucrose, which is translocated up to the growing shoot, so the root has become a source and the shoot is a sink.

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## 20 · Part 1 Extended Translocation [6]

**Mark scheme**

**(a)** **[2]**

- sucrose or amino acids are translocated (down) from the leaves (source) in the phloem (1)
- they cannot pass the ring because the phloem is removed, so they build up above the ring (1)

**ALLOW** the build-up of food above the ring is used for growth there

**(b)** **[2]**

- the xylem is still intact or undamaged (1)
- so water (and mineral ions) can still be transported up to the leaves (1)

**(c)** **[2]**

- no sucrose or amino acids can reach the roots (because the phloem is cut) (1)
- roots need sucrose for respiration or growth, so once their stores run out they die (1)

**Worked solution**

(a) Leaves make sucrose and send it, with amino acids, down the stem in the phloem. Removing the ring of bark removes the phloem, so these substances cannot move past the ring. They build up just above it, and the extra supply makes the stem swell there. (b) The xylem was not damaged, so water and mineral ions still travel up from the roots to the leaves. The leaves keep their water supply and do not wilt. (c) The roots are sinks: they cannot photosynthesise and rely on sucrose from the leaves for respiration. With the phloem cut, no sucrose reaches them, so once their own stores are used up the roots die.

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